

ISS Greenhouse Module

Campaign 1 · Case 01 · Low Earth Orbit



SOLAR
AGRICULTURAL
AGENCY
Teacher Guide

PREPARATION

Before class

- Confirm the game loads.
- Print the student sheet double-sided or use fillable HTML mode.
- Keep the evidence fallback ready.
- Provide headphones.

GAME ACCESS

Launch path

interstellarharvest.github.io/Space-Sprout-Sleuth/

Select Campaign 1 → Case 1.

CORRECT DIAGNOSIS

Central interpretation

Microgravity disrupts gravitropism. Roots need an alternative consistent cue or guidance system.

ESSENTIAL EVIDENCE

Evidence students should notice

Source	Essential evidence
Crew	Roots do not grow down; resource adjustments did not solve the pattern.
Sensors	Microgravity is the major unusual condition; nutrients, water, temperature, humidity, and lighting are nominal.
Plants	Dense tangles, poor anchoring, and unusual stem angles.
Logs	Gravitropism/statolith mechanism and possible replacement cues.

LIKELY STICKING POINT

Microgravity is not the literal absence of gravity. The ISS and everything inside it are continuously falling around Earth, producing apparent weightlessness.

COLLECTION

What to collect

3 · Investigate four evidence sources · 6 · Diagnose and reject an alternative · 7 · Claim-Evidence-Reasoning · 8 · Supply a consistent orientation cue · 9 · Exit ticket

LESSON FLOW

60-minute plan

Time	Action
0–5	Launch game and materials
5–10	Initial prediction
10–30	3 · Investigate four evidence sources
30–38	4 · Test the competing explanations
38–48	6 · Diagnose and reject an alternative + 7 · Claim-Evidence-Reasoning
48–55	8 · Supply a consistent orientation cue
55–60	9 · Exit ticket

FALLBACK

Technical fallback

Use the evidence digest on the final teacher page. Students complete the same academic task without the game.

TEACHER LINE

“Your job is not to click everything. Your job is to determine which evidence best explains the root pattern.”

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OVERVIEW

Lesson overview

Students investigate a stunted lettuce crop aboard the International Space Station. They compare crew testimony, environmental sensors, direct plant observations, and mission logs to determine why roots form tangled balls despite normal resources. Students explain the gravitropism mechanism and evaluate an engineered alternative cue.

GUIDING QUESTION

Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

ALIGNMENT

Standards alignment

- **MS-ETS1-1:** Supports defining criteria and constraints for a root-guidance solution.
- **MS-ETS1-2:** Supports evaluating competing orientation solutions.
- **SEP:** Constructing Explanations and Designing Solutions; Analyzing and Interpreting Data.
- **DCI:** ETS1.A and ETS1.B, with plant structure/function as supporting science.
- **CCC:** Cause and Effect; Structure and Function.

This lesson supports components of the standards; it does not by itself fulfill an entire performance expectation.

SUCCESS CRITERIA

What proficient work includes

Relevant evidence from at least three sources · correct mechanism · evidence-based alternative rejection · concise CER · directional engineering response.

REFERENCE

Vocabulary

gravitropism · microgravity · statocyte · statolith · phototropism

OUTCOMES

Learning objectives

1. Distinguish observations from explanations across four evidence sources.
2. Identify disrupted gravitropism as the best-supported cause.
3. Explain statolith movement as part of a directional gravity-sensing mechanism.
4. Reject an alternative explanation using evidence.
5. Define a criterion and constraint for an orientation system.

PREPARATION

Materials

Chromebooks · headphones · game access · double-sided mission sheet or editable HTML · evidence fallback.

NOTES

PREPARATION

Teacher preparation

1. Open the game and confirm Campaign 1, Case 1 loads.
2. Test audio/headphones.
3. Print or share the student sheet.
4. Keep the evidence fallback ready.

☒ PROCEDURE

0–5 minutes · Setup

Students launch the game and open the mission sheet. State that game score, rank, speed, and optional dialogue are not academically graded.

PROCEDURE

5–10 minutes · Phenomenon launch

Show the greenhouse image or read the briefing. Ask what evidence would distinguish a resource problem from an orientation problem. Students complete **2 · Initial thinking**; do not confirm the diagnosis.

PROCEDURE

10–30 minutes · Gameplay investigation

Students visit crew, sensors, plants, and logs. They record only evidence that supports or rejects a cause.

FACILITATION PROMPTS

Which detail is a symptom? Which normal reading rules out an explanation? Have you used more than one source?

☒ PROCEDURE

30–38 minutes · Evidence checkpoint

Students complete **4 · Test the competing explanations**. Confirm that resource readings are normal and that the root pattern is directional and repeated. Do not require optional dialogue completion.

PROCEDURE

38–48 minutes · Diagnosis and CER

Students complete **6 · Diagnose and reject an alternative** after the game unlocks, then complete **7 · Claim-Evidence-Reasoning** using evidence from several sources and reasoning that links microgravity to a weakened directional cue.

PROCEDURE

48–55 minutes · Engineering application

Students choose an alternative cue and give one criterion and constraint. Strong responses distinguish a direct root cue from lighting that primarily guides shoots.

☒ PROCEDURE

55–60 minutes · 9 · Exit ticket

Students answer independently. Collect or save the mission file.

✓ FORMATIVE CHECKS

Checks for understanding

- Separates symptom from cause.
- Uses normal readings to reject alternatives.
- Describes microgravity as apparent weightlessness.
- Links statolith direction to root orientation.
- Proposes a cue that addresses the problem.

📄 ASSESSMENT

Evidence of learning

Formative: prediction, evidence matrix, explanation ranking, teacher questions.

Collected: diagnosis, rejected alternative, CER, engineering response, exit ticket.

♿ ACCESS

Accessibility supports

- Bullet-point evidence accepted.
- CER frame available but optional.
- Text-to-speech compatible.
- Game evidence may remain open.
- Evidence fallback available.
- No group product required.

❓ MISCONCEPTIONS

Common misconceptions

1. **No gravity:** The ISS remains under gravity while continuously falling around Earth.
2. **Roots use only gravity:** Roots also respond to moisture, touch, chemicals, and other cues.
3. **Brighter light fixes roots:** Light is a stronger shoot cue; roots need a more direct guide.
4. **Plants cannot grow in space:** Growth is possible with adapted environmental systems.
5. **Plant pillows recreate gravity:** They manage media and fluids, not a gravity vector.

🧪 SOURCE STATUS

Science status



ESTABLISHED

gravitropism, statolith involvement, phototropism, plant responses to microgravity.



PLAUSIBLE ENGINEERING

root channels, moisture cues, rotating systems. Students complete **8 · Supply a consistent orientation cue** on the worksheet; Campaign 1 has no in-game apply-the-fix step, so these solutions appear only as discussion in the mission logs, not as a selectable action.



FICTIONAL CONTEXT

the specific emergency, crew, SAA, and exact symptoms.

🔑 FALLBACK

Technical fallback

Distribute the evidence digest. Students complete the same mission sheet; technical failure is not graded.

NOTES

EVIDENCE ARCHITECTURE Evidence map

Source	Essential evidence	Role
Crew	Roots do not grow down; changing resources did not help.	Symptom and failed interventions
Sensors	Microgravity; normal nutrients, water, climate, and lighting.	Strong causal difference and elimination evidence
Plants	Tangled roots, poor anchoring, unusual stems.	Direct biological observation
Logs	Gravitropism/statolith mechanism and possible replacement cues.	Mechanism and design options

REASONING PATH How the case resolves

1. Root tangling is the symptom.
2. Normal resources weaken simple resource explanations.
3. Microgravity is the major unusual environmental condition.
4. Logs connect gravity sensing to directional growth.
5. Disrupted gravitropism best explains the pattern.
6. A useful solution supplies another consistent cue.

DISTRACTORS Competing explanations

Distractor	Why tempting	Why rejected
Nutrient burn	Pale, stunted plants	Nominal solution; directional tangling is a poor match
Wrong light cycle	Odd stems and pale leaves	16/8 and PAR are nominal; roots are the strongest abnormality
Defective seeds	All plants affected	Pattern repeats with environment and prior plantings
Water problem	Roots seek water	Delivery works; uniform wicking may reduce a cue but does not best explain the case

The nutrient, light, and seed distractors are the three incorrect options students can select on the game's Diagnose screen. "Water problem" is an additional teacher discussion distractor for class conversation — it is not one of the selectable in-game diagnoses.

BOUNDARY NOTE

The game simplifies a complex plant response. Do not teach that roots are completely random, that statoliths are the only gravity-sensing mechanism, or that plant pillows fully solve orientation.

INSTRUCTIONAL IDENTITY Student role and primary practice

Student role: Pattern investigator. **Primary practice:** compare multiple evidence types, eliminate alternatives, explain a mechanism, and define an engineering problem.

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☒ CHECKLIST RUBRIC

Quick classroom rubric

Dimension	Secure	Developing	Beginning
Evidence	Specific, relevant evidence from several sources	Some relevant evidence; important detail/source missing	Vague, copied, or irrelevant evidence
Mechanism	Connects microgravity, statolith direction, and root orientation	Names gravitropism but mechanism is incomplete	Cause stated without mechanism
Reasoning	Connects evidence to claim and rejects an alternative	Partial connection or weak rejection	Evidence listed without explanation
Communication	Accurate vocabulary and complete product	Minor vocabulary/completion gaps	Major gaps impede meaning

ANALYTIC GRID

Formal analytic rubric

Criterion	4 · Accomplished	3 · Proficient	2 · Developing	1 · Beginning
Evidence selection	Most relevant; symptom/cause distinguished	Relevant with minor omissions	Mixed relevant/irrelevant	Little usable evidence
Source diversity	Integrates 3–4 source types	Uses at least 2 source types	Mostly one source	Sources unclear
Mechanism	Accurate and appropriately qualified	Central mechanism accurate	Important gaps	Unsupported/inaccurate
Alternative evaluation	Specific contradictory evidence	Generally relevant evidence	Weak support	No evaluation
Engineering	Feasible cue, criterion, constraint	Relevant solution + criterion/constraint	Loosely related solution	Does not address orientation
Communication	Clear, concise, complete	Mostly clear/complete	Inconsistent but understandable	Incomplete/difficult

SCORING BOUNDARY

No numeric points appear on the student page by default. Game score, rank, speed, and optional exploration are never academic criteria.

NOTES

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AUTHORITATIVE SOURCES

Source list

1. NASA, *Growing Plants in Space*
nasa.gov/exploration-research-and-technology/growing-plants-in-space/
2. NASA Ames, *Plant Gravity Perception (Space-13)*
nasa.gov/ames/space-biosciences/plant-gravity-perception-spacex-13/
3. NASA, *Veggie Plant Growth System Activated on ISS*
nasa.gov/missions/station/veggie-plant-growth-system-activated-on-international-space-station/
4. NASA Science, *Plant Biology Program · Hardware*
science.nasa.gov/biological-physical/focus-areas/plant-biology/hardware/
5. Ditengou, Teale & Palme (2020), *Settling for Less: Do Statoliths Modulate Gravity Perception?*
pubmed.ncbi.nlm.nih.gov/31963631/
6. NASA OSDR, *Arabidopsis Seedlings in Microgravity (OSD-38)*
osdr.nasa.gov/bio/repo/data/studies/OSD-38
7. NGSS MS-ETS1-1 and MS-ETS1-2
nextgenscience.org

EVIDENCE FALLBACK

No-game evidence digest

Briefing: ISS lettuce is stunted; roots curl around plant pillows instead of anchoring.

Crew: roots do not grow downward; changing resources did not help.

Sensors: microgravity; nutrients, water, temperature, humidity, and lighting are nominal.

Plants: dense tangles, poor anchoring, and angled stems.

Logs: statolith-based gravity sensing normally provides direction; possible compensations include root channels, moisture gradients, shoot lighting, or rotating systems.

TECHNICAL NOTES

Classroom fallback

- Refresh once if the game does not load.
- Use a current Chromebook Chrome tab.
- The Diagnose screen unlocks after formal clues are collected.
- Do not require every optional conversation branch.
- Move to the fallback within two minutes rather than losing lesson time.

NOTES